

ITA 0608 - MACHINE LEARNING

3 . Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Aim

To demonstrate the working of the Decision Tree based ID3 (Iterative Dichotomiser 3) algorithm by constructing a decision tree using a sample dataset and using the generated tree to classify a new sample.

Algorithm

- Start with the complete training dataset.
- Calculate the entropy of the dataset to measure impurity.
- Compute the Information Gain (IG) for each attribute.
- Select the attribute with the highest Information Gain as the root node of the decision tree.
- Split the dataset into subsets based on the values of the selected attribute.
- Repeat Steps 2–5 recursively for each subset using the remaining attributes.
- Stop recursion when:
 - All samples in a subset belong to the same class, or
 - No attributes remain for further splitting, or
 - The subset is empty (assign the majority class).
- Assign class labels to all leaf nodes.
- Use the constructed decision tree to classify new or unseen samples by traversing from the root to the appropriate leaf node.
- End.

Code

```
new_sample = pd.DataFrame([{'Outlook': 'Sunny',  
    'Temperature': 'Cool',  
    'Humidity': 'High',  
    'Wind': 'Strong'  
}])  
print("New Sample to Classify:")  
display(new_sample)  
le_outlook = LabelEncoder().fit(data['Outlook'])  
le_temp = LabelEncoder().fit(data['Temperature'])  
le_humidity = LabelEncoder().fit(data['Humidity'])  
le_wind = LabelEncoder().fit(data['Wind'])  
new_sample['Outlook'] = le_outlook.transform(new_sample['Outlook'])  
new_sample['Temperature'] = le_temp.transform(new_sample['Temperature'])  
new_sample['Humidity'] = le_humidity.transform(new_sample['Humidity'])  
new_sample['Wind'] = le_wind.transform(new_sample['Wind'])  
prediction = model.predict(new_sample)  
le_play_tennis = LabelEncoder().fit(data['PlayTennis'])  
predicted_class = le_play_tennis.inverse_transform(prediction)  
print(f"\nPrediction for the new sample: {predicted_class[0]}")
```

Output

*** New Sample to Classify:

	Outlook	Temperature	Humidity	Wind
0	Sunny	Cool	High	Strong

Prediction for the new sample: No

Result

The ID3 Decision Tree algorithm was successfully implemented using the given dataset. The decision tree was constructed by selecting attributes based on the highest Information Gain, with Outlook chosen as the root node.